

3D-POS: 3D Pareto Optimized Head Selection for Fog-enabled Smart EEG Healthcare IoT

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Abstract—This paper proposes a master head selection framework for fog-enabled EEG-based smart Healthcare Internet of Things (HIoT) systems. In a smart healthcare environment where multiple fog devices handle multi-channel EEG data, selecting an optimal fog device as the processing head is critical to ensure low delay, efficient energy consumption, and reliable data handling. Given the heterogeneous nature of fog devices in real-world settings, we formulate the head selection as a multi-objective optimization problem, considering the key factors such as delay, energy, computation capacity, and proximity to EEG devices. By employing Pareto optimization techniques, the framework identifies a set of non-dominated head candidates among all the fog devices that achieve the best trade-offs among conflicting objectives across various scenarios. The simulation results demonstrate that the proposed approach significantly minimizes 10 – 50% delay compared to existing methods under varying workloads and different configurations of EEG and fog devices. This work enhances the responsiveness and sustainability of fog-enabled IoT infrastructures by intelligently balancing delay, energy consumption, and computational efficiency.

Index Terms—Electroencephalogram (EEG), Healthcare Internet of Things (HIoT), Fog Computing, Pareto Optimization, Master Head Selection

I. INTRODUCTION

Over the past decade, fog computing has emerged as a highly beneficial paradigm for the Healthcare Internet of Things (HIoT). The fog computing brings computational services closer to the healthcare devices compared to the cloud computing to healthcare IoT devices, facilitating real-time processing of healthcare data [1]. While cloud computing services supports intensive processing, big healthcare data management and long-term data storage, they raise certain challenges such as higher bandwidth requirements, higher cost and a centralized architecture [2]. The HIoT network comprises healthcare devices like Electroencephalogram (EEG), Electrocardiogram (ECG), Electrooculogram (EOG), Electromyogram (EMG), etc., which generates signals from various parts of human body and analyze them using computing resources. This paper focuses on the fog-enabled services for processing EEG data within the HIoT network. The EEG signals, collected from the scalp of the human brain, are one of the most critical biomedical signals that play a key role in brain disease diagnosis, human-computer interaction, emotion and sentiment analysis, virtual reality gaming and other Artificial Intelligence (AI) driven applications [3] [4]. Since EEG healthcare data is time-sensitive, real-time processing with minimal latency is essential for healthcare applications. To meet these requirements, fog computing is preferred over cloud computing, as it offers lower latency, lower cost, faster responses, and decentralized

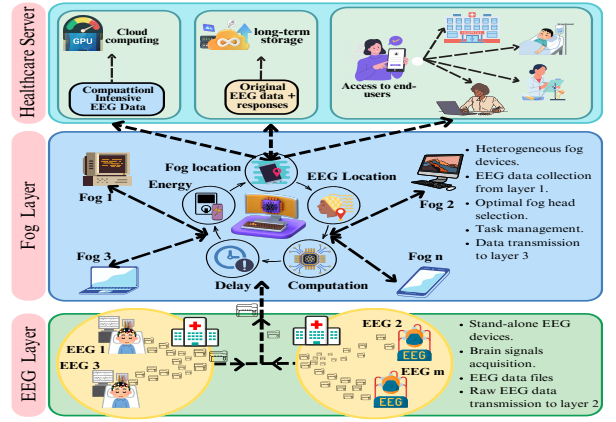


Fig. 1: Shows the layered architecture of proposed 3D-POS.

processing [5] [6] [7]. The fog computing devices typically have limited processing power and storage capacity [8]; on the other side, the EEG devices generate continuous brain signals, resulting in large volumes of data [9]. A single fog device often cannot handle such data alone, necessitating the use of multiple heterogeneous fog devices within the HIoT network. Therefore, it is crucial to optimally select a head among the fog devices that is responsible for collecting EEG data from the EEG devices and manage task distribution across the network. This work contributes the optimal head selection by considering various parameters related to EEG data, fog devices, and the network in the HIoT environment.

Motivation and contributions: EEG devices generate a continuous high volume of data that requires low-latency processing [10]. Making real-time data analysis in the healthcare servers introduce additional delays, which is unacceptable for critical applications. Fog computing reduces the need to send large EEG data streams to the server, saving bandwidth and latency. Further, Fog nodes have limited energy, storage and processing capabilities [11] [12]. Continuous processing of EEG tasks may lead to rapid depletion. Therefore there is a need of multiple Fog devices in the HIoT network, but they vary in processing power and energy capacity. To address the above challenges, we propose an optimal head selection for EEG data management in a multi-EEG multi-fog smart HIoT distributed network. The key objectives are -

- We design our framework for heterogeneous fog network, it is essential to account for the variability across multiple parameters. To address this, we optimize key factors including latency, energy consumption, and computational

efficiency to enable the selection of the most suitable fog device as the master head.

- Considering heterogeneity, simple heuristic approaches often fail to achieve a balanced trade-off among conflicting objectives. To overcome this, we employ a three-dimensional Pareto-optimized selection strategy that dynamically identifies the most efficient fog device.
- This work provides solutions for all possible optimization cases including delay-minimized, energy-optimized, performance-maximized, as well as balanced trade-off solutions among all three objectives. We evaluate the proposed framework using benchmark EEG datasets across multiple scenarios and conduct performance and comparative analyses.

II. SYSTEM ARCHITECTURE AND NETWORK DESIGN

This work proposes a parameter-based fog head selection strategy, referred to as 3D-POS, aims to select an optimal fog device using three-dimensional Pareto optimization technique. Fig. 1 depicts the three-layered system architecture for our work. Layer 1 includes multiple stand-alone EEG devices that acquire brain signals and transmit to layer 2 as EEG data files. Layer 2 contains heterogeneous fog devices and an optimal head that collects the EEG data files to further distribute among multiple fog devices. Moreover, the optimal head is connected to a healthcare server for transmission of EEG data and responses. The healthcare server in layer 3 is utilized for long-term storage, intensive computing, and access to the end users.

A. Network Design

The HIoT network comprises a set of EEG devices and fog devices as follows- Let $\mathbb{E}$ represent the set of EEG devices, given by $\mathbb{E} = \{e_1, e_2, \dots, e_m\}$, $e \in \mathbb{E}$, and $e_i \not\leftrightarrow e_j$; where $e_i \not\leftrightarrow e_j$ indicates that two EEG devices are independent in nature. similarly, let $\mathbb{F} = \{f_1, f_2, \dots, f_n\}$, $f_i \in \mathbb{F}$, and $f_i \leftrightarrow f_j$; where $f_i \leftrightarrow f_j$ indicates that fog devices are interconnected. Let an i^{th} EEG device generate a set of tasks $\mathbb{T}^{e_i}$, the size of these tasks is denoted as $\mathcal{S}(\mathbb{T})^{e_i}$.

Definition 1: The Device Proximity Factor (DPF) of a fog device is defined as the ratio of the summation of EEG Proximity Factor (EPF) and Fog Proximity Factor (FPF) to the Centrality Factor (CF).

For the given set of coordinates (x_{f_j}, y_{f_j}) for j^{th} fog device and (x_{e_i}, y_{e_i}) for i^{th} EEG device, the EPF of j^{th} fog device is calculated as the average Euclidean distance between the fog device and the EEG devices. Mathematically, it is given by:

$$EPF_j = \frac{1}{m} \sum_{i=1}^m \sqrt{(x_{f_j} - x_{e_i})^2 + (y_{f_j} - y_{e_i})^2}, \quad (1)$$

$\forall e_i \in \mathbb{E}$ and $\forall f_j \in \mathbb{F}$. Similarly, the FPF of j^{th} fog device is calculated as the average Euclidean distance between the fog device and the rest of fog devices in the network. Mathematically, it is given by:

$$FPF_j = \frac{1}{n-1} \sum_{\substack{j=1 \\ j \neq \bar{j}}}^{n-1} \sqrt{(x_{f_j} - x_{f_{\bar{j}}})^2 + (y_{f_j} - y_{f_{\bar{j}}})^2}, \quad (2)$$

$\forall f_j, f_{\bar{j}} \in \mathbb{F}$. The CP of j^{th} fog device is computed as the sum of the total Euclidean distances from (i) the fog device to all EEG devices and (ii) the fog device to rest of the fog devices. Therefore,

$$CF_j = \sum_{i=1}^m \sqrt{(x_{f_j} - x_{e_i})^2 + (y_{f_j} - y_{e_i})^2} + \sum_{\substack{j=1 \\ j \neq \bar{j}}}^n \sqrt{(x_{f_j} - x_{f_{\bar{j}}})^2 + (y_{f_j} - y_{f_{\bar{j}}})^2} \quad (3)$$

Finally, Eq. (4) is derived from Eq. (1), (2), and (3) as follows,

$$DPF_j = \frac{EPF_j + FPF_j}{CF_j} \quad (4)$$

Definition 2: The Device Energy Efficiency (DEE) of a fog device is defined as the normalized residual energy of the device. It is calculated as the ratio of the difference between the Device Total Energy (DTE) and the sum of Device Consumption Energy (DCE), Device Operational Energy (DOE) to the DTE.

The fog device consumes energy during EEG data transmission and receiving operations in the HIoT network, referred to as DOE. Assuming, e_j^t as the unit energy consumption by the fog device during data transmission and receiving, the DOE of j^{th} fog device is calculated as,

$$DOE_j = e_j^t \times \frac{\sum_{i=1}^m \mathcal{S}(\mathbb{T})^{e_i}}{\mathcal{B}} \quad (5)$$

where, $\mathcal{B}$ represents the bandwidth of the network. Let the unit energy consumption by j^{th} fog device during processing be noted as e_j^c . The processing energy consumption by the fog device is calculated as,

$$DCE_j = e_j^c \times \frac{\sum_{i=1}^m \mathcal{S}(\mathbb{T})^{e_i}}{\mathcal{C}_j^{\text{base}}} \quad (6)$$

where, $\mathcal{C}_j^{\text{base}}$ is the base clock frequency of j^{th} fog device. Finally, the DEE_j is calculated from Eq. (5) and (6) as,

$$DEE_j = \frac{DTE_j - (DOE_j + DCE_j)}{DTE_j} \quad (7)$$

Definition 3: The Device Computation Capability (DCC) of a fog device is measured by the relative tradeoff between Cycles Per Instruction (CPI) and Memory Utilization Factor (MUF).

The CPI count of j^{th} fog device depends on the following factors - number of cores (θ_j), average clock frequency ($\mathcal{C}_j^{\text{avg}}$), and million instructions per second ($\mathcal{M}_j$). The CPI is calculated as follows:

$$CPI_j = \frac{\theta_j \times \mathcal{C}_j^{\text{avg}}}{\mathcal{M}_j \times 10^6} \quad (8)$$

where, $\mathcal{C}_j^{\text{avg}}$ is calculated as $\frac{\mathcal{C}_j^{\text{base}} + \mathcal{C}_j^{\text{max}}}{2}$. Further, MUF of j^{th} fog device is calculated as follows:

$$MUF_j = \frac{\Delta M_j^{\text{avail}} - \Delta M_j^{\text{use}}}{\Delta M_j^{\text{avail}}} \quad (9)$$

In Eq. (9), $\Delta M_j^{\text{avail}}$ represents the available primary memory at a given time, while ΔM_j^{use} denotes the occupied primary

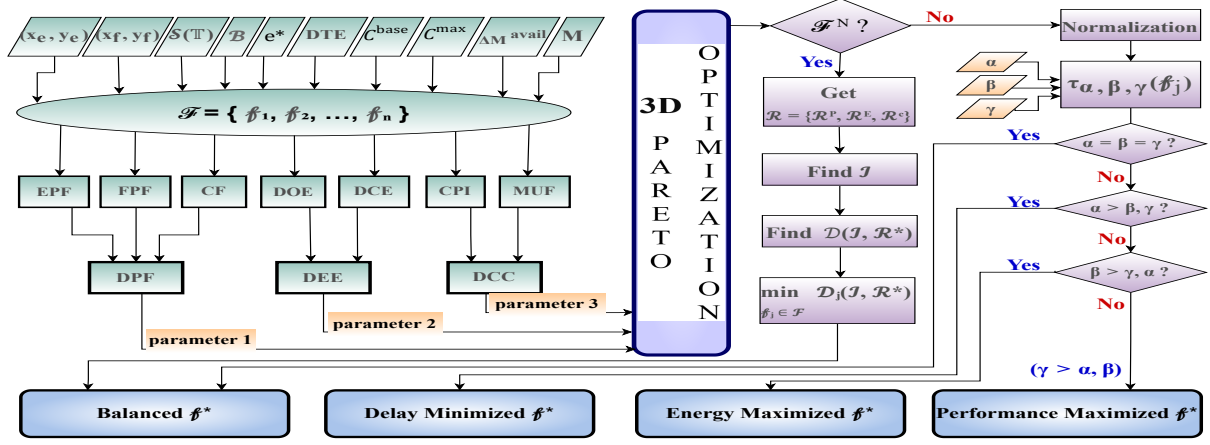


Fig. 2: Describes the detailed workflow of 3D-POS in three parts - inputs to the decision parameters, multi-objective pareto optimization steps, final results for all possible scenarios.

memory. Finally, DCC_j is calculated by the weighted combination of CPI_j and MUF_j using Eq. (8) and (9) as follows:

$$DCC_j = \alpha \left(\frac{\theta_j \times C_j^{\text{avg}}}{M_j \times 10^6} \right) + \beta \left(\frac{\Delta M_j^{\text{avail}} - \Delta M_j^{\text{use}}}{\Delta M_j^{\text{avail}}} \right) \quad (10)$$

Here, α and β are the scaling factors that adjust the relative importance of CPI and MUF in determining the DCC.

III. PROPOSED OPTIMAL HEAD SELECTION

A. Problem Formulation

Our optimal head selection strategy is based on the three decision parameters - (i) DPF, (ii) DEE, and (iii) DCC. The goal is to select the master head among the heterogeneous fog devices in the IoT network that achieves the best trade-off among minimum DPF and maximum DEE, DCC. We define our objective function with the optimal fog device f^* as,

$$f^* = \arg \min_{f_j \in \mathbb{F}} (\alpha \cdot \text{DPF}_j - \beta \cdot \text{DEE}_j - \gamma \cdot \text{DCC}_j) \quad (11)$$

subject to:

$$C1: \mathbb{F}^N = \{f_j \in \mathbb{F} \mid \nexists f_k \in \mathbb{F} \text{ such that } f_k \succ f_j\}$$

$$C2: \mathcal{D}(f^*, f_b^*) \leq \mathcal{D}^{\text{max}}, \quad \exists f_b^* \in F$$

$$C3: \text{DTE}_j \geq e^{\text{min}}$$

Each fog device $f_j \in \mathbb{F}$ is associated with a feature vector $\mathbb{V} = (\text{DPF}_j, \text{DEE}_j, \text{DCC}_j)$ considering parameter weights as α, β, γ respectively. C1 explains there exists a non-dominated solution space $\mathbb{F}^N \in \mathbb{F}$ where f_j dominates f_k is denoted as $f_j \succ f_k$. C2 deals with the proximity constraint where f^* assumes to have a backup device f_b^* within a maximum threshold distance $\mathcal{D}^{\text{max}}$. C3 ensures the f^* must have a minimum DTE. Since DPF is to minimize and DEE, DCC are to maximize, there is least possible optimal solution that simultaneously optimizes all three criteria perfectly. Therefore, we re-transform our problem to a multi-objective Pareto based optimization problem, assuming $\mathcal{R}$ be the decision variable, given in Eq. (12), (13) and (14). The decision value for an objective is obtained by solving:

$$\mathcal{R}^P = \arg \min_{f_j \in \mathbb{F}^N} \mathcal{R}^P(f_j) \quad (12)$$

$$\mathcal{R}^E = \arg \max_{f_j \in \mathbb{F}^N} \mathcal{R}^E(f_j) \quad (13)$$

$$\mathcal{R}^C = \arg \max_{f_j \in \mathbb{F}^N} \mathcal{R}^C(f_j) \quad (14)$$

In this problem, there are multiple uncertainty cases - (i) If a fog device has the lowest DPF, it may not have the highest DEE or DCC. (ii) If a fog device has the highest DEE or DCC, it may not have the lowest DPF. In this case, the best solution is the trade-off between the parameter values. The Pareto optimization problem seeks to identify solutions where improving one parameter (treated as objective) cannot be done without worsening at least one other objective.

B. Solution Approach

In the optimal head selection process, since minimizing DPF conflicts with maximizing DEE and DCC, a single best solution may not exist. Instead, we use Pareto optimization to find the set of non-dominated solutions, ensuring a fair and justified selection of the optimal fog device from a set of feasible solutions. We consider a few assumptions as follows:

- **Assumption 1:** A fog device $f_j \in \mathbb{F}$ is said to dominate another fog device $f_k \in \mathbb{F}$, denoted as $f_j \succ f_k$ if $\text{DPF}_j \leq \text{DPF}_k$, $\text{DEE}_j \geq \text{DEE}_k$, $\text{DCC}_j \geq \text{DCC}_k$.
- **Assumption 2:** With at least one strict inequality as $\text{DPF}_j < \text{DPF}_k$ or $\text{DEE}_j > \text{DEE}_k$ or $\text{DCC}_j > \text{DCC}_k$.
- **Assumption 3:** An optimal fog device is one that is not dominated by any other fog device, referring to C1 of Eq. 11.

The workflow of the proposed optimal head selection is given in Fig. 2.

Step-1: Each fog device in the IoT network calculates the following parameter values - DPF using Eq. (4), DEE using Eq. (7), and DCC using Eq. (10).

Step-2: The fog devices are categorized into either Non-dominated Solution Space, $\mathbb{F}^N$, or Dominated Solution Space, $\mathbb{F}^D$, where $\mathbb{F}^N \cup \mathbb{F}^D = \mathbb{F}$ and $\mathbb{F}^N \cap \mathbb{F}^D = \phi$. The $\mathbb{F}^D$ (in Eq. (16)) contains all fog devices that are inferior to at least one other fog device in the network. The $\mathbb{F}^N$ (in Eq. (15))

represents trade-off solutions where no single objective can be improved without worsening another.

$$\mathbb{F}^N = \{f_j \in \mathbb{F} \mid \nexists f_k \in \mathbb{F} \text{ such that } f_k \succ f_j\} \quad (15)$$

$$\mathbb{F}^D = \{f_j \in \mathbb{F} \mid \exists f_k \in \mathbb{F} \text{ such that } f_j \succ f_k\} \quad (16)$$

Step-3a: For the exist $\mathbb{F}^N$, a pareto front $\mathcal{P}(\mathbb{F}^N)$ is established having a set of decision values as given in Eq. (17), denoted as Reference Points $\mathcal{R} = \{\mathcal{R}^P, \mathcal{R}^E, \mathcal{R}^C\}$ from $\downarrow$ DPF, $\uparrow$ DEE, and $\uparrow$ DCC respectively. The reference points are the extreme solutions that optimize each individual objective independently, without considering trade-offs between objectives.

$$\mathcal{P}(\mathbb{F}^N) = \{\mathcal{R}(f_j) \mid f_j \in \mathbb{F}^N\} \quad (17)$$

Step-3b: If $\mathbb{F}^N = \phi$, then the optimal head selection is based on the tradeoff function $\tau_{\alpha,\beta,\gamma}(f_j)$ between three parameters defined in Eq. (18).

$$\begin{aligned} \tau_{\alpha,\beta,\gamma}(f_j) = & \alpha \times \frac{\text{DPF}_j}{\sum_{j=1}^n \text{DPF}_j} + \beta \times \frac{\text{DEE}_j}{\sum_{j=1}^n \text{DEE}_j} \\ & + \gamma \times \frac{\text{DCC}_j}{\sum_{j=1}^n \text{DCC}_j} \end{aligned} \quad (18)$$

Step-4: From the reference points, an Ideal Point ($\mathcal{I}$) is observed that represents the best possible values for all objectives simultaneously, assuming no trade-offs exist. Basically, $\mathcal{I}$ is a hypothetical point and serves as a reference for real solutions that are to the best possible outcome. For example, if $\mathcal{R}^P = a$, $\mathcal{R}^E = b$, and $\mathcal{R}^C = c$, then the position of $\mathcal{I}$ in the 3D representation is (a, b, c) .

Step-5: Finally, f^* is selected from $\mathcal{F}^N$ considering any of these three scenarios listed below:

- *Scenario 1:* If $|\mathbb{F}^N| = 1$, then the only fog device is selected as the optimal head and the back up fog device is selected as described in *Step-3b*.
- *Scenario 2:* If $|\mathbb{F}^N| > 1$, then the optimal fog head and the backup fog device are selected based on the minimal distance from $\mathcal{I}$ to $\mathcal{R}^*$ using Eq. (19) as,

$$\mathcal{D}(\mathcal{I}, \mathcal{R}^*) = \sqrt{(x_{\mathcal{I}} - x_{\mathcal{R}^*})^2 + (y_{\mathcal{I}} - y_{\mathcal{R}^*})^2}, \quad (19)$$

$$\forall \mathcal{R}^* \in \{\mathcal{R}^P, \mathcal{R}^E, \mathcal{R}^C\}$$

- *Scenario 3:* If $|\mathbb{F}^N| = 0$, then again there are four possible cases:
 - If $\alpha > \beta, \gamma$; solution is delay minimized.
 - If $\beta > \gamma, \alpha$; solution is energy maximized.
 - If $\gamma > \alpha, \beta$; solution is performance maximized.
 - If $\alpha = \beta = \gamma$; solution is balanced.

Illustrative Example: Let us consider an environment having 8 heterogeneous fog devices and 3 EEG devices randomly distributed across the network. We take a cluster-like EEG-fog placement randomly generated as shown in Fig. 3. The optimal master head is f_8 that is the only fog device in $\mathbb{F}^N$, as mentioned in *Scenario 1*. Whereas Fig. 4 depicts the selection of master head *i.e.* f_5 in a more distributed device placement from the set of non-dominated solutions $\mathbb{F}^N = \{f_5, f_7, f_8\}$, as given in *Scenario 2*.

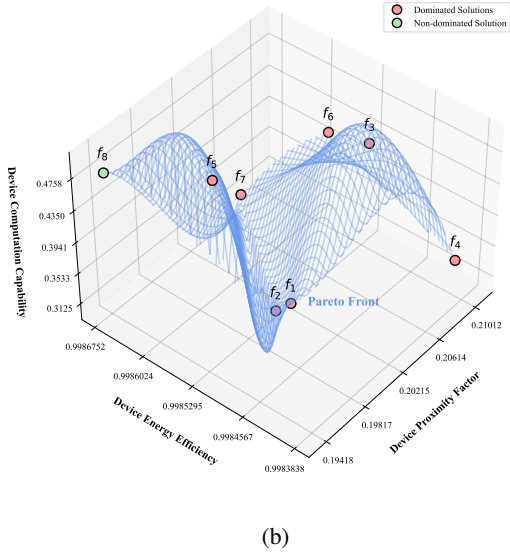
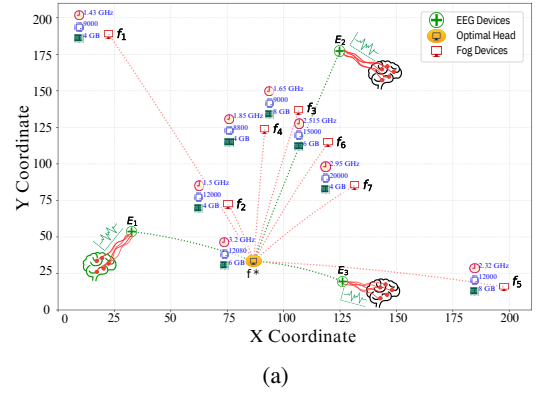


Fig. 3: Shows the implementation of 3D-POS for *Scenario 1*, where $|\mathbb{F}^N| = 1$. The only non-dominated solution is selected as the master head. (a) Shows the random placement of EEG and fog devices in a cluster-like distribution. (b) Depicts $\mathbb{F}^N$ and $\mathbb{F}^D$ along with the pareto front for all three parameters.

IV. EXPERIMENTAL RESULT ANALYSIS

We set up the simulation environment using Python v3.12 in 16 GB RAM, i7 processor considering the simulation area of maximum 200×200 units. For this experiment, we have considered the fog dataset given in [13] and benchmark EEG datasets [14] [15]. We have shown the performance of 3D-POS by varying number of devices and tasks considering more than hundred iterations in each analysis. In our experiment, we have taken the number of fog devices $|\mathbb{F}| = [2, 10]$, EEG devices $|\mathbb{E}| = [1, 3]$ and a single healthcare server with $\mathcal{B} = 500$ Mbps. We compared our model with other existing models [13] and [16].

We analyze the average total delay (in seconds) as a key performance evaluation metric. Fig. 5 presents the relationship between the number of tasks and average total delay (in seconds) across different fog configurations. The configurations with more fog devices experience lower delay compared to configurations with fewer fog nodes suggesting that a higher number of fog nodes helps in better load distribution.

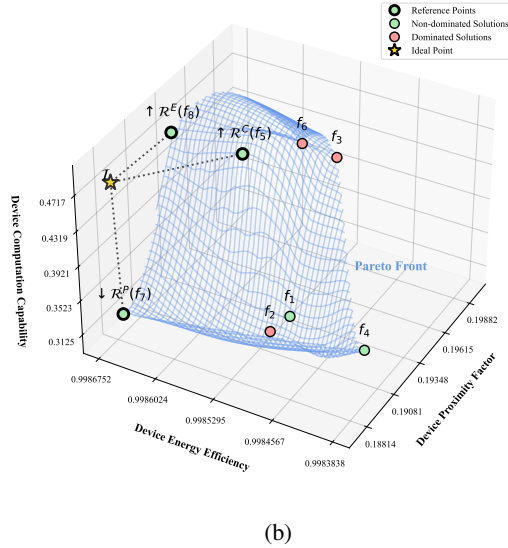
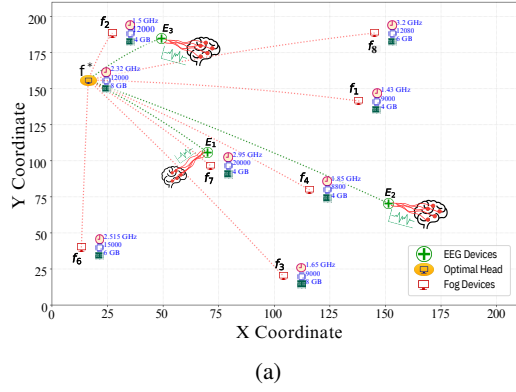


Fig. 4: Shows the implementation of 3D-POS for *Scenario 2*, where $|\mathbb{F}^N| > 1$. The optimal master head is selected based on the minimal $\mathcal{D}(\mathcal{I}, \mathcal{R}^*)$. (a) Shows the random placement of EEG and fog devices in a more distributed environment. (b) Depicts $\mathbb{F}^N$, $\mathbb{F}^D$, and the pareto front highlighting the reference points and the ideal point.

Therefore, as the number of tasks increases (from 10 to 100), the average total delay (minimum 0.66 s to maximum 6.62 s) also increases for all configurations.

Additionally, We consider the average processing delay (in second) and Device Computation Capability (DCC) as performance metrics showing the relationship with increasing number of fog devices and two different clock frequencies - base clock frequency ($\mathcal{C}^{\text{base}}$) and boosting clock frequency ($\mathcal{C}^{\text{max}}$). As the number of fog devices increases, the average processing delay decreases and there is a steady change in DCC, shown in Fig. 6. The trend suggests that a higher number of fog devices leads to lower processing delays. Moreover, the Boosting Clock Frequency shows higher computation capability and lower delay compared to the base Clock Frequency indicates that the fog devices process tasks faster and reduce delay, operating at higher clock speeds. Whereas, the overall computation capability can steadily increase or decrease with the change in number of fog devices due to the randomness

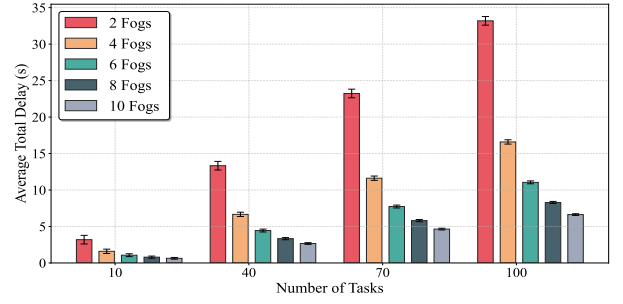


Fig. 5: Shows variations of average total delay with the number of tasks for different fog node configurations.

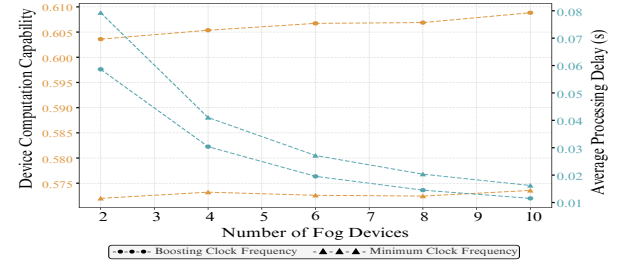


Fig. 6: Shows the performance analysis of DCC along with average performance delay for different clock frequencies while change in number of fog devices.

of device selection.

Further, Fig. 7 examines the relationship between the number of tasks and average energy consumption (in Joules) for different fog node configurations. The average energy consumption rises with an increase in the number of tasks. As processing more tasks requires additional computational effort, leading to higher power consumption. For 2 fog nodes, the energy consumption starts low and increases steadily. For 10 fog nodes, the energy consumption is significantly higher even at a low task count, but the rate of increase is more gradual. This indicates that in smaller fog networks, energy consumption increases drastically when tasks exceed the available computational capacity.

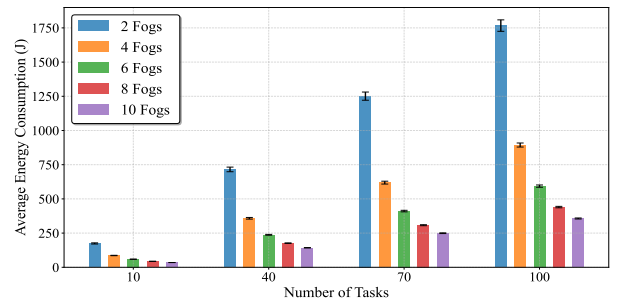


Fig. 7: Shows the average energy consumption with the number of tasks for different fog node configurations illustrating the impact of task distribution on energy usage.

The results depicted in Fig. 8 compare the average total delay between the proposed 3D-POS method and the existing

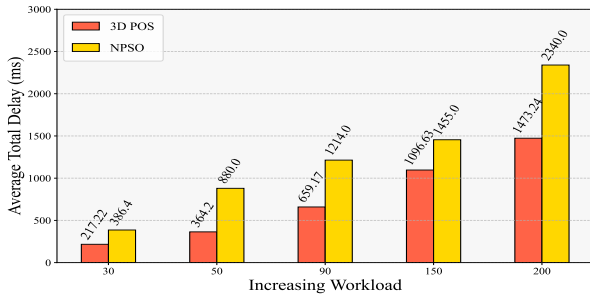


Fig. 8: Shows comparative analysis between proposed 3D-POS and existing NPSO methods for increasing workload and corresponding total delay.

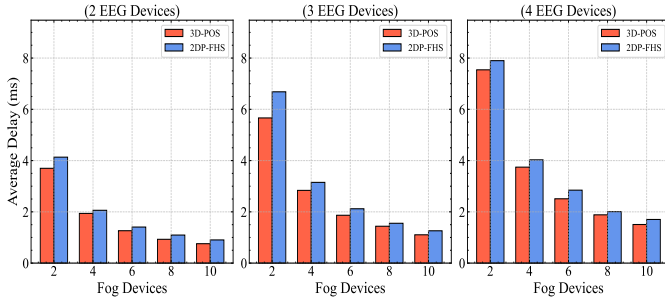


Fig. 9: Shows average delay comparison between proposed 3D-POS and existing 2DP-FHS methods for an increasing number of fog devices for multiple EEG devices.

NPSO method [16] across varying workload levels. 3D-POS performs slightly better at lower workloads (30 and 50). At moderate workloads (90 and 150), the performance gap widens significantly. At high workload (200), delay for 3D-POS is 1473.2 units, whereas NPSO reaches a very high 2300.0 units. This shows that 3D-POS is more scalable and can handle higher workloads much better in terms of delay.

Furthermore, Fig. 9 highlights the comparison of average delay between the proposed 3D-POS and the existing 2DP-FHS method [13]. The evaluation is conducted across three different EEG device settings considering multiple fog devices. We observe that 3D-POS consistently achieves lower delay compared to 2DP-FHS across all EEG-fog configurations. The percentage reduction in delay ranges from 5.88% to 16.24% for 2 EEG devices, from 7.44% to 15.25% for 3 EEG devices, and from 4.52% to 11.81% for 4 EEG devices. It is also observed that the trend in average delay shows a variable increasing or decreasing pattern due to the inherent randomness across iterations.

V. CONCLUSION

This work proposes the optimal head selection among a number of fog devices in the HIoT network consisting of EEG healthcare devices. The selection of master head includes all key decision factors such as delay, energy, and computational efficiency including EEG, fog, and network-related parameters that affect in decision-making. We adopted a multi-objective Pareto optimization technique for master head selection considering all possible scenarios. In comparative analysis of 3D-

POS with recent existing works, we get that our proposed work reduces total delay of average 41.88% and 10.41% compared to NPSO and 2DP-FHS, showing the effectiveness in EEG-fog-IoT applications. In the future work, we aim to develop an efficient task distribution mechanism for this work considering multi-EEG multi-fog IoT framework. Further, we would implement a cost efficient model for the same work that is best suitable for small-scale EEG-based healthcare centers.

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